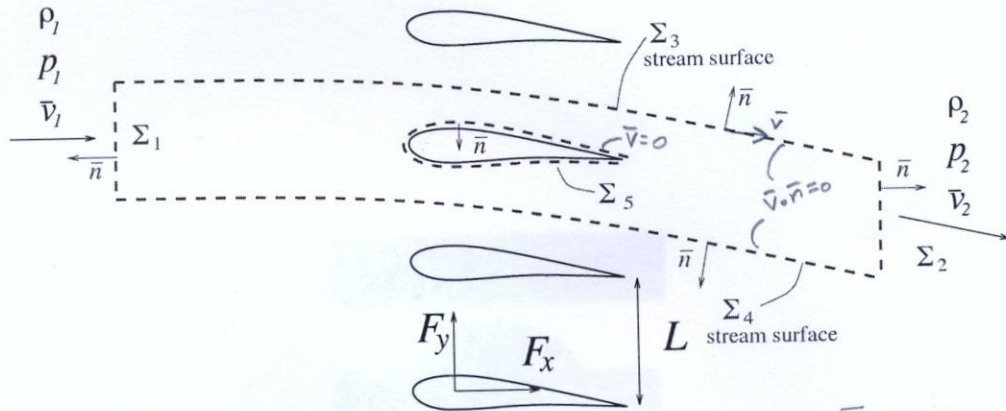


CL4

A planar uniform stream of air with pressure p_1 , density ρ_1 and velocity $\bar{v}_1 = v_{1x}\bar{e}_x$ flows past an array of aerofoils with vertical separation L , exerting on each one a force (per unit length) given by $\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y$. Assuming that the aerofoils are thermally insulated and neglecting the effect of gravity, write the equations that determine the values of the pressure p_2 , density ρ_2 and velocity $\bar{v}_2 = v_{2x}\bar{e}_x + v_{2y}\bar{e}_y$ for the uniform stream found sufficiently far downstream.



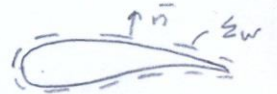
CONT: $\frac{d}{dt} \int_{V_c} \rho dV + \int_{\Sigma_c} \rho(\bar{v} - \bar{v}_c) \cdot \bar{n} d\sigma = 0 \Rightarrow \rho_1 v_{1x} L = \rho_2 v_{2x} L$

$\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4 + \Sigma_5$

$\bar{v} \cdot \bar{n} = 0$ on $\Sigma_3, \Sigma_4, \Sigma_5$

$\boxed{\rho_1 v_{1x} = \rho_2 v_{2x}}$

$$\bar{F} = - \int_{\Sigma_w} p \bar{n} d\sigma + \int_{\Sigma_w} \bar{e} \cdot \bar{n} d\sigma$$



MOMENTUM: $\frac{d}{dt} \int_{V_c} \rho \bar{v} dV + \int_{\Sigma_c} \rho \bar{v} [(\bar{v} - \bar{v}_c) \cdot \bar{n}] d\sigma = - \int_{\Sigma_c} p \bar{n} d\sigma + \int_{\Sigma_c} \bar{e} \cdot \bar{n} d\sigma + \int_{V_c} \rho \bar{g} dV$

$- \rho_1 v_{1x} v_{1x} \bar{e}_x + \rho_2 v_{2x} [v_{2x} \bar{e}_x + v_{2y} \bar{e}_y] L = p_1 L \bar{e}_x - p_2 L \bar{e}_x - \int_{\Sigma_s} p \bar{n} d\sigma + \int_{\Sigma_s} \bar{e} \cdot \bar{n} d\sigma$

$\underbrace{\rho_1 v_{1x}}_{\rho_1 v_{1x}}$

$\underbrace{- \int_{\Sigma_s} p \bar{n} d\sigma + \int_{\Sigma_s} \bar{e} \cdot \bar{n} d\sigma}_{-\bar{F}}$

$\boxed{\frac{F_x}{L} = p_1 - p_2 + \rho_1 v_{1x} (v_{1x} - v_{2x})}$

$\boxed{\frac{F_y}{L} = - \rho_1 v_{1x} v_{2y}}$

ENERGY: $\frac{d}{dt} \int_{V_c} \rho \left(e + \frac{v^2}{2} \right) dV + \int_{\Sigma_c} \rho \left(e + \frac{v^2}{2} \right) [(\bar{v} - \bar{v}_c) \cdot \bar{n}] d\sigma = - \int_{\Sigma_c} p \bar{v} \cdot \bar{n} d\sigma + \int_{\Sigma_c} \bar{v} \cdot \bar{e} \cdot \bar{n} d\sigma + \int_{V_c} \rho \bar{v} \cdot \bar{g} dV + \int_{V_c} k \nabla T \cdot \bar{n} d\sigma + \int_{V_c} (\dot{Q}_e + \dot{Q}_r) dV$

$\Sigma_1 + \Sigma_2 + \Sigma_3 + \Sigma_4 + \Sigma_5$

$\bar{v} \cdot \bar{n} = 0$ on $\Sigma_3, \Sigma_4, \Sigma_5$

$\bar{v} = 0$ on Σ_5

$\bar{v} = 0$ on Σ_3, Σ_4

$\nabla T = 0$

CANCEL EACH OTHER OUT

CANCEL EACH OTHER OUT

THERMALLY INSULATED

$\int_{\Sigma_1 + \Sigma_2} \rho \left(e + \frac{p}{\rho} + \frac{v^2}{2} \right) \bar{v} \cdot \bar{n} d\sigma = 0 \Rightarrow \rho_1 v_{1x} L \left[\left(h_2 + \frac{v_{2x}^2 + v_{2y}^2}{2} \right) - \left(h_1 + \frac{v_{1x}^2}{2} \right) \right] = 0$

$h = \frac{\gamma}{\gamma-1} \frac{p}{\rho} \Rightarrow \boxed{\frac{\gamma}{\gamma-1} \frac{p_2}{\rho_2} + \frac{v_{2x}^2 + v_{2y}^2}{2} = \frac{\gamma}{\gamma-1} \frac{p_1}{\rho_1} + \frac{v_{1x}^2}{2}}$